

Cambridge (CIE) IGCSE Chemistry
Paper 4: Theory (Set F)
Extended

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 80

Instructions

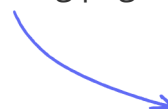
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

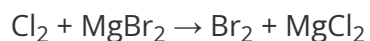
- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



1 (a) Aqueous chlorine reacts with aqueous magnesium bromide.



How does this reaction show that chlorine is more reactive than bromine?

(1 mark)

(b) What colour is bromine in aqueous solution?

(1 mark)

(c) Complete the word equation to show the halogen and halide compound which react to form the products iodine and sodium chloride.



(2 marks)

(d) Explain, in terms of the reactivity of the halogens, why aqueous bromine will not react with aqueous sodium chloride.

(1 mark)

2 (a) Element Z is in Period 3 and Group V. Identify element Z.

.....
(1 mark)

(b) Helium, neon and argon are noble gases.

Explain, in terms of the electronic structure, why neon is unreactive.

.....
(1 mark)

(c) Copper is a transition element. It can have variable oxidation states.

State two other chemical properties of transition elements which make them different from Group I elements.

.....
.....
(2 marks)

3 (a) Soluble salts can be made using a base and an acid.

Complete this method of preparing dry crystals of the soluble salt cobalt(II) chloride-6-water from the insoluble base cobalt(II) carbonate.

step 1 Add an excess of cobalt(II) carbonate to hot dilute hydrochloric acid. **step 2**

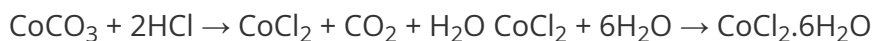
step 3

step 4

(4 marks)

(b) i)

5.95 g of cobalt(II) carbonate were added to 40 cm³ of hydrochloric acid, concentration 2.0 mol / dm³. Calculate the maximum yield of cobalt(II) chloride-6-water and show that the cobalt(II) carbonate was in excess.



maximum yield:

number of moles of HCl used =

number of moles of CoCl₂ formed =

number of moles of CoCl₂·6H₂O formed =

mass of one mole of CoCl₂·6H₂O = 238 g

maximum yield of CoCl₂·6H₂O =g

to show that cobalt(II) carbonate is in excess:

number of moles of HCl used = (use your value from above)

mass of one mole of CoCO₃ = 119 g

number of moles of CoCO₃ in 5.95 g of cobalt(II) carbonate =

[5]

ii) Explain how these calculations show that cobalt(II) carbonate is in excess.

[1]

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(6 marks)

4 (a) This question is about elements **X**, **Y** and **Z**.

An atom of element **X** is represented as ${}^{34}_{16}\text{X}$

Name the different types of particles found in the nucleus of this atom of **X**.

.....
.....
(2 marks)

(b) What is the term for the total number of particles in the nucleus of an atom?

.....
(1 mark)

(c) What is the total number of particles in the nucleus of an atom of ${}^{34}_{16}\text{X}$?

.....
(1 mark)

(d) What is the electronic structure of the ion X^{2-} ?

.....
(1 mark)

(e) What term is used to describe atoms of the same element with different numbers of particles in the nucleus?

.....
(1 mark)

- (f) Part of the definition of relative atomic mass is 'the average mass of naturally occurring atoms of an element'. Some relative atomic masses are not whole numbers. Element Y has only two different types of atom, ^{69}Y and ^{71}Y . The ratio of atoms present in element Y is shown.

$$^{69}\text{Y} : ^{71}\text{Y} = 3 : 2$$

- Calculate the relative atomic mass of element Y to one decimal place.

relative atomic mass:

- Identify element Y.
- element Y:

.....

.....

.....

(3 marks)

- (g) Element Z is in Period 3 and Group V.

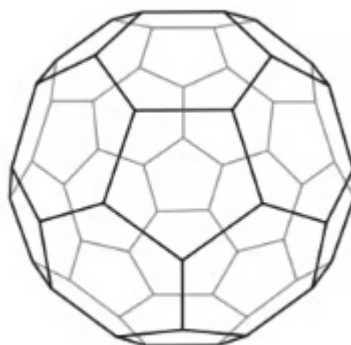
Explain in terms of electron transfer why Z behaves chemically as a non-metal.

.....

.....

(2 marks)

- 5 (a)** In 1985 the fullerenes were discovered. They are solid forms of the element carbon. The structure of the C_{60} fullerene is given below.



- i) In the C_{60} fullerene, how many other carbon atoms is each carbon atom bonded to?

[1]

- ii) Another fullerene has a relative molecular mass of 840. How many carbon atoms are there in one molecule of this fullerene?

[1]

(2 marks)

- (b)** Fullerenes are soluble in liquid hydrocarbons such as octane. The other solid forms of carbon are insoluble.

Describe how you could obtain crystals of fullerenes from soot which is a mixture of fullerenes and other solid forms of carbon.

(3 marks)

(c) A mixture of a fullerene and potassium is an excellent conductor of electricity.

i) Which other form of solid carbon is a good conductor of electricity?

[1]

ii) Explain why metals, such as potassium, are good conductors of electricity.

[2]

iii) The mixture of fullerene and potassium has to be stored out of contact with air. There are substances in unpolluted air which will react with potassium. Name **two** potassium compounds which could be formed when potassium is exposed to air.

[2]

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(5 marks)

6 (a) Two homologous series of hydrocarbons are the alkanes and the alkenes.

One general characteristic of a homologous series is that the physical properties vary in a predictable way.

i) State **three** other general characteristics of a homologous series.

[3]

ii) How can the molecular formula of a hydrocarbon show whether it is an alkane or an alkene?

[2]

iii) How do alkanes and alkenes differ in their molecular structures?

[2]

(7 marks)

(b) Cracking is the thermal decomposition of alkanes into smaller hydrocarbons and possibly hydrogen.

i) State **two** conditions required for the cracking of an alkane.

[2]

ii) One type of cracking produces an alkane and an alkene. Complete an equation for the cracking of heptane into an alkane and an alkene.



[1]

iii) Complete an equation for the cracking of heptane into hydrogen and two other products.



[1]

iv) Suggest **one** reason why cracking is important.

[1]

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(5 marks)

(c) Hydrocarbons burn in excess oxygen to form carbon dioxide and water. 20 cm^3 of a gaseous hydrocarbon burned in an excess of oxygen, 200 cm^3 . After cooling, the volume of the residual gas at r.t.p. was 150 cm^3 , 50 cm^3 of which was oxygen.

i) Determine the volume of the oxygen used.

[1]

ii) Determine the volume of the carbon dioxide formed.

[1]

.....
.....
(2 marks)

7 (a) Monomers polymerise to form polymers or macromolecules.

i) Explain the term *polymerise*.

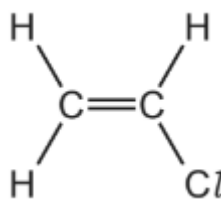
[1]

ii) There are two types of polymerisation - addition and condensation. What is the difference between them?

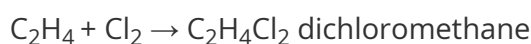
[2]

(3 marks)

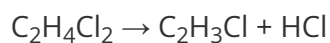
(b) An important monomer is chloroethene which has the structural formula shown below.



It is made by the following method.



This is heated to make chloroethene.

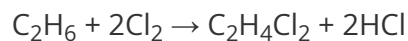


i) Ethene is made by cracking alkanes. Complete the equation for cracking dodecane.



[1]

Another method of making dichloroethane is from ethane.



ii) Suggest a reason why the method using ethene is preferred.

[1]

iii) Describe an industrial method of making chlorine.

[2]

iv) Draw the structural formula of poly(chloroethene).

Include three monomer units.

[2]

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(6 marks)

- 8 (a)** An ore of tungsten contains WO_3 . Tungsten can be obtained from WO_3 by reacting it with hydrogen. The reaction is a redox reaction, water is also produced.

Write the balanced symbol equation for the reaction.

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(2 marks)

- (b)** Explain reduction and oxidation in terms of electrons and oxidation numbers.

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.....

(2 marks)

- (c)** The oxidation number of oxygen is -2 and this remains unchanged during the reaction.

Identify, using oxidation numbers, what has been oxidised and what has been reduced in this reaction.

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(3 marks)

- (d)** 50 tonnes of the tungsten ore were processed. The ore was found to contain 3.4% of WO_3 by mass. Calculate the maximum mass, in tonnes, of tungsten that could be obtained from this ore.

Give your answer to **three** significant figures.

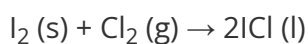
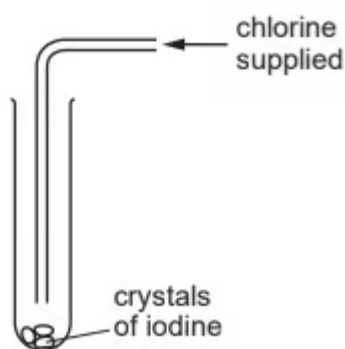
$[A_r(\text{W}) = 184; \quad M_r(\text{WO}_3) = 232]$

maximum mass of tungsten = tonnes

.....

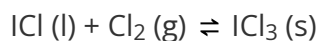
(5 marks)

9 (a) Iodine reacts with chlorine to form a dark brown liquid, iodine monochloride.



dark brown

When more chlorine is added and the tube is sealed, a reversible reaction occurs and the reaction comes to equilibrium.



dark brown

yellow

i) Give another example of a reversible reaction.

[1]

ii) Explain the term *equilibrium*.

[2]

(3 marks)

(b) Chlorine is removed from the tube and a new equilibrium is formed.

Explain why there is less of the yellow solid and more dark brown liquid in the new equilibrium mixture.

(2 marks)